

Building scalable systems with microservices

Task :

Your engineering team has been contracted to architect the core backend for *UrbanRide*, an emerging **global** ride-hailing platform. The platform must process **1 million completed rides per day**, maintain **sub-second latency** (< 500ms) for high-frequency transactions, absorb sudden traffic surges during peak demand (e.g., adverse weather or major public events), and operate on a cost-optimized cloud infrastructure budget. (maintaining cost per completed ride below 10 LKR)

Technical System Requirements

- **Microservices Architecture:** Domain-driven service separation (e.g., Trip Management, Driver Location Ingestion, Matching Engine, Surge Pricing, Billing & Notifications).
- **High Throughput & Spikes:** Event-driven streaming mechanisms (e.g., Apache Kafka, RabbitMQ) to decouple heavy write traffic (continuous GPS pings) from real-time ride-matching workflows.
- **Low-Latency Geospatial Operations:** In-memory caching structures (e.g., Redis Geospatial, Uber H3, or Google S2 indexes) ensuring sub-second matching response times.
- **Cost Efficiency Strategy:** Cost-aware compute topologies (e.g., spot/preemptible instances for stateless microservices, dynamic auto-scaling groups, serverless execution for low-frequency paths, and tiered data archiving).

Required Deliverables

1. **Architecture Design Document (ADD)**
 - **Domain & Service Decomposition:** Microservices boundaries, inter-service communication protocols (gRPC vs. REST vs. WebSockets).
 - **Visual Architecture Diagrams:** Standard C4 Model diagrams (Context, Container, Component levels).
 - **Scalability & Resiliency Blueprint:** Circuit breakers, rate limiters, backpressure strategies, and dead-letter queues to handle burst traffic without system failure.
 - **Infrastructure & Cost Breakdown:** Estimated resource usage, instance sizing rationale, auto-scaling triggers, and storage tiering strategy.
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2. Presentation Deck

- Visual slide deck formatted strictly for a 10-minute technical architectural review.

Evaluation & Defense Setup

- **Presentation (10 Minutes):** The group presents the high-level architecture, trade-off decisions, and cost-optimization strategies.
- **Panel Q&A (10 Minutes):** An industry expert panel tests the design against real-world edge cases (e.g., database lock contention, network partitions, geo-index cache invalidation, and sudden 5x traffic spikes).

Grading Rubric

Category	Weight	Key Assessment Focus
Microservice & Data Design	25%	Clear bounded contexts, database isolation, distributed transaction handling (Saga pattern).
Performance & Latency	25%	Sub-second geospatial querying, caching layers, write-heavy ingestion handling.
Cost Optimization	20%	Auto-scaling design, spot instance utilization, data tiering, resource footprint balance.
Presentation Clarity	15%	Standardized C4 diagrams, time management (10-minute adherence), technical delivery.
Panel Defense (Q&A)	15%	Reasoning under panel questioning, fault tolerance mechanisms, CAP theorem trade-offs.

Submission Due Date

Midnight, Sunday 13th September 2026